

LLM-инженер

online course ✨

модуль #6

лекция: разработка приложений

Архитектуры чат-ботов

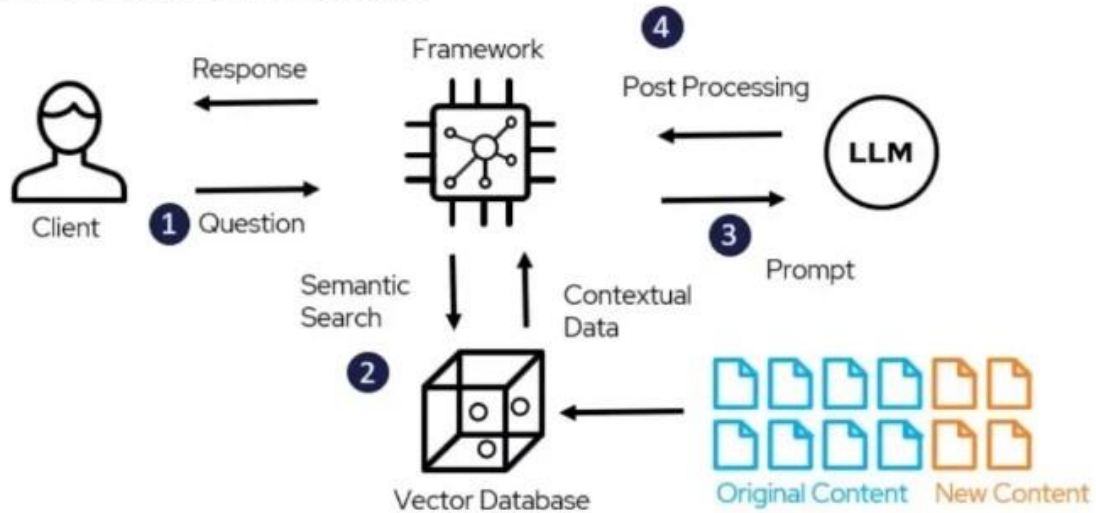
Правила

How A Rule-Based Chatbot Works

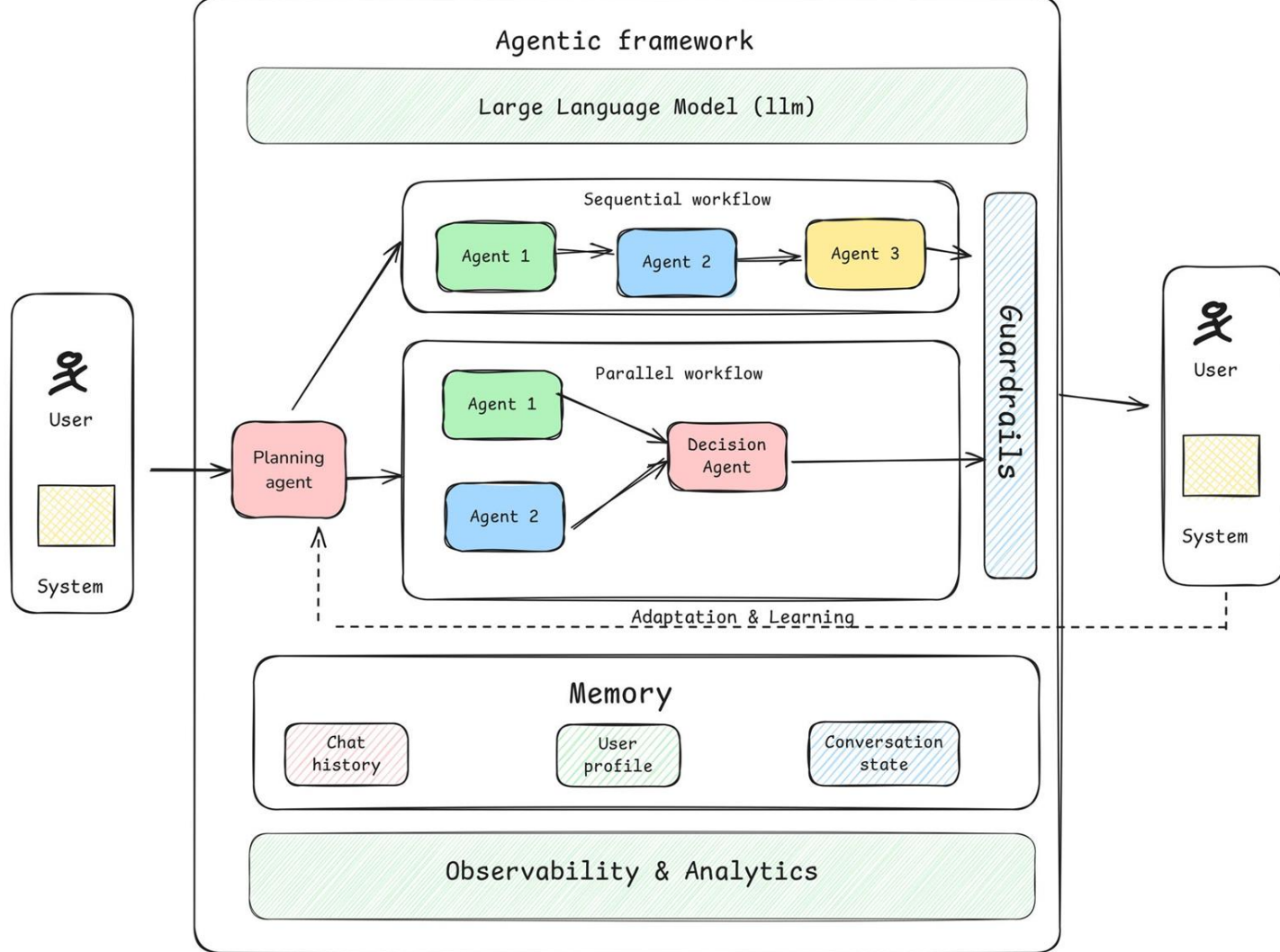


RAG

RAG Architecture Model



Agents



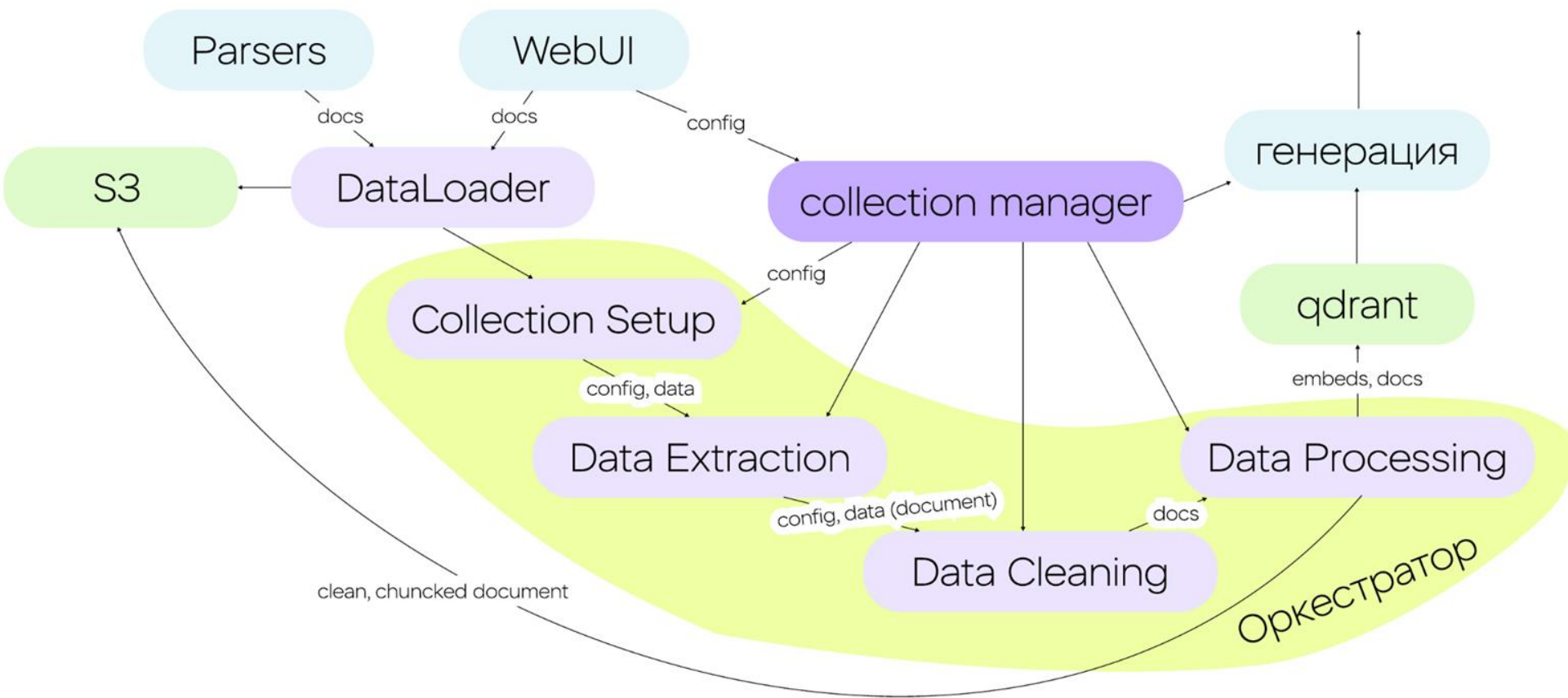
парсинг статей

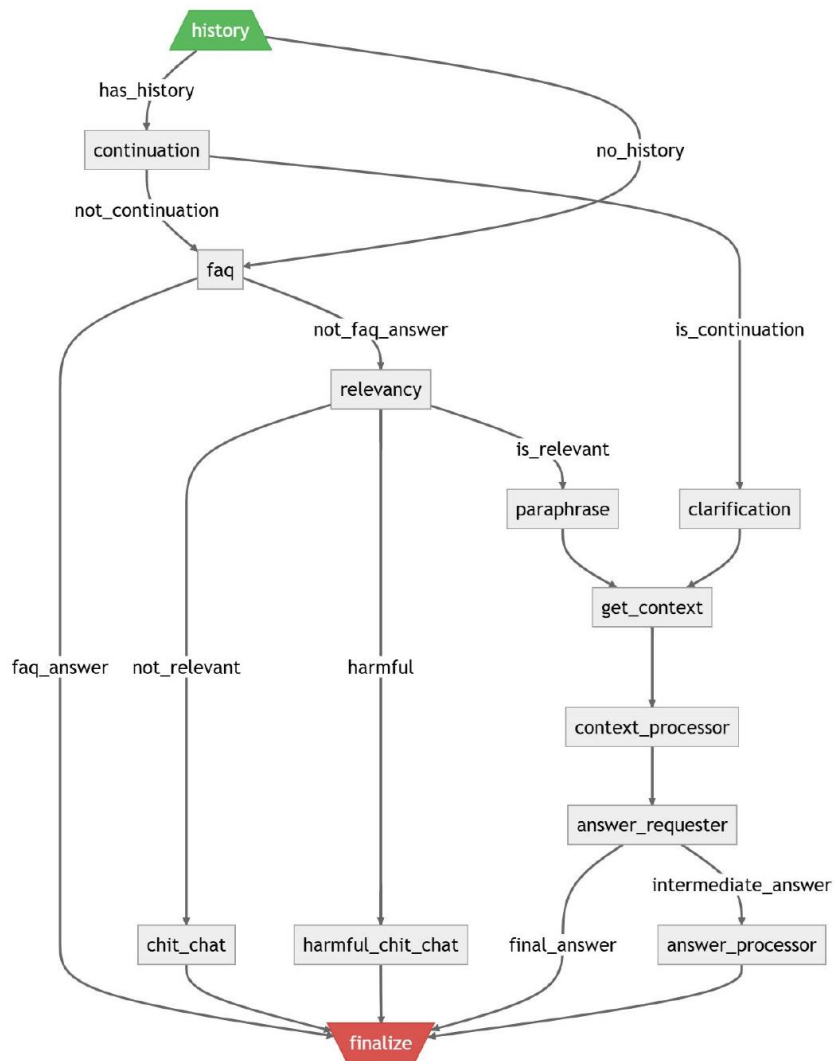
пары вопрос-ответ

ретривер

гибко настраиваемый пайплайн
генерации ответа







LLM (general)

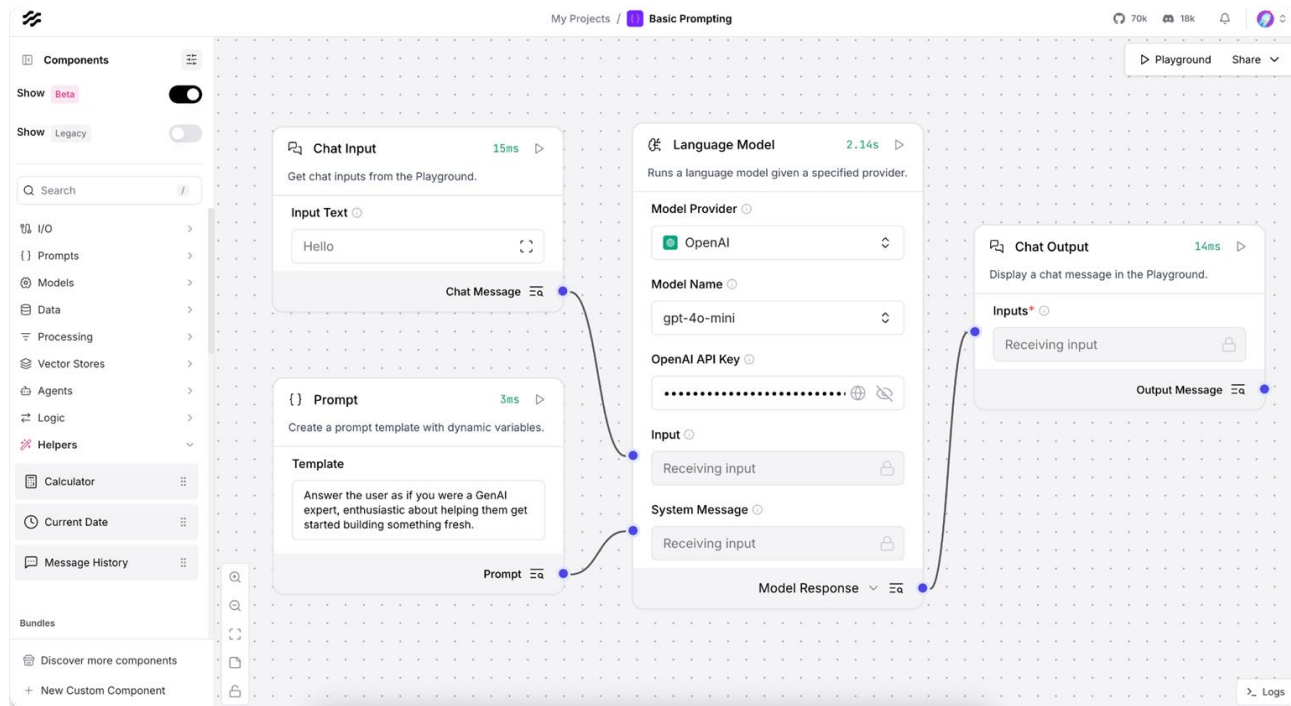
- OpenAI
- LangChain
- LlamaIndex

Agentic

- [Pydantic.AI](#)
- Swarm
- LangGraph
- ...

Low code

- Langflow
- n8n



Вспомогательные инструменты

Пакетные менеджеры

- pip
- poetry
- uv
- pdm

```
[dependency-groups]
lint = [
    "ruff==0.14.0", # locked so we don't get PRs which fail only due to a lint update
    "pre-commit>=4.1.0",
    "bibtextparser>=1.4.3", # used for tests/test_citation_formatting.py
    "typos>1",
]
test = [
    "pytest>=8.3.4,<8.4.0",
    "pytest-xdist>=3.6.1,<3.7.0",
    "pytest-coverage>=0.0",
    "pytest-rerunfailures>=15.0,<16.0",
    "iso639>=0.1.4", # used for tests/scripts/test_generate_model_meta.py
    "GitPython>=3.0.0",
]
```

```
[project]
name = "mteb"
version = "2.3.0"
description = "Massive Text Embedding Benchmark"
readme = "README.md"
authors = [
    { name = "MTEB Contributors", email = "niklas@huggingface.co" },
    { name = "Kenneth Enevoldsen", email = "kenneth.enevoldsen@cas.au.dk" },
    { name = "Nouamane Tazi", email = "nouamane@huggingface.co" },
    { name = "Nils Reimers", email = "info@nils-reimers.de" },
]
maintainers = [
    { name = "Kenneth Enevoldsen", email = "kenneth.enevoldsen@cas.au.dk" },
    { name = "Roman Solomatin", email = "risolomatin@gmail.com" },
    { name = "Isaac Chung", email = "chungisaac1217@gmail.com" },
]
license = "Apache-2.0"
license-files = ["LICENSE"]
keywords = ["deep learning", "text embeddings", "embeddings", "multimodal", "benchmark", "retrieval", "information retrieval"]
classifiers = [
    "Development Status :: 5 - Production/Stable",
    "Environment :: Console",
    "Intended Audience :: Developers",
    "Intended Audience :: Information Technology",
    "Operating System :: OS Independent",
    "Programming Language :: Python",
]
requires-python = ">=3.10,<3.15"
dependencies = [
    "datasets>=2.19.0",
    "numpy>=1.0.0,<3.0.0",
    "requests>=2.26.0",
    "scikit-learn>=1.4.0",
    "scipy>=0.0.0",
    "sentence_transformers>=3.0.0",
    "typing_extensions>=4.5.0",
    "torch>1.0.0",
    "tqdm>1.0.0",
    "rich>=0.0.0",
    "pytorch-eval-terrier>=0.5.0",
    "pydantic>=2.0.0",
    "polars>=0.20.22",
]
```

Линтеры & формatters

- Black
- Isort
- Flake8
- ruff

Тyпecek

- Mypy
- ty

```
[tool.ruff.lint]
select = [
    "F", # pyflakes rules,
    "I", # sorting for imports
    "E", # formatting for docs
    "D", # formatting for docs
    "UP", # upgrade to latest syntax if possible
    "FA", # Future annotations
    "C4", # cleaner comprehensions
    "NPY", # numpy
    "N", # PEP8 naming convention
    "RUF009", # function-call-in-dataclass-default-argument
    "RUF008", # mutable-dataclass-default
    "RUF013", # implicit-optional
    "RUF012", # mutable-class-default
    "RUF015", # unnecessary-iterable-allocation-for-first-element
    "RUF016", # invalid-index-type
    "RUF017", # quadratic-list-summation
    "RUF018", # assignment-in-assert
    "RUF019", # unnecessary-key-check
    "RUF020", # never-union
    "RUF021", # parenthesize-chained-operators
    "RUF022", # unsorted-dunder-all
    "RUF024", # mutable-fromkeys-value
    "RUF026", # default-factory-kwarg
    "RUF033", # post-init-default
    "RUF034", # useless-if-else
    "RUF041", # unnecessary-nested-literal
    "RUF040", # invalid-assert-message-literal-argument
    "RUF046", # unnecessary-cast-to-int
    "RUF051", # if-key-in-dict-del
    "RUF100", # unused-noqa
    "RUF101", # redirected-noqa
    "RUF200", # invalid-pyproject-toml
    "PTH", # use pathlib
    "TID", # tidy-imports
    "D", # pydocstyle
]
```

Деплоймент

Из скриптов

- + Легко проверить что не так
- + Удобно дебажить
- Не воспроизводимо между окружениями

```
name: deploy
on:
  push:
    branches:
      - main
  workflow_dispatch:

jobs:
  run_pull:
    name: run pull
    runs-on: ubuntu-latest

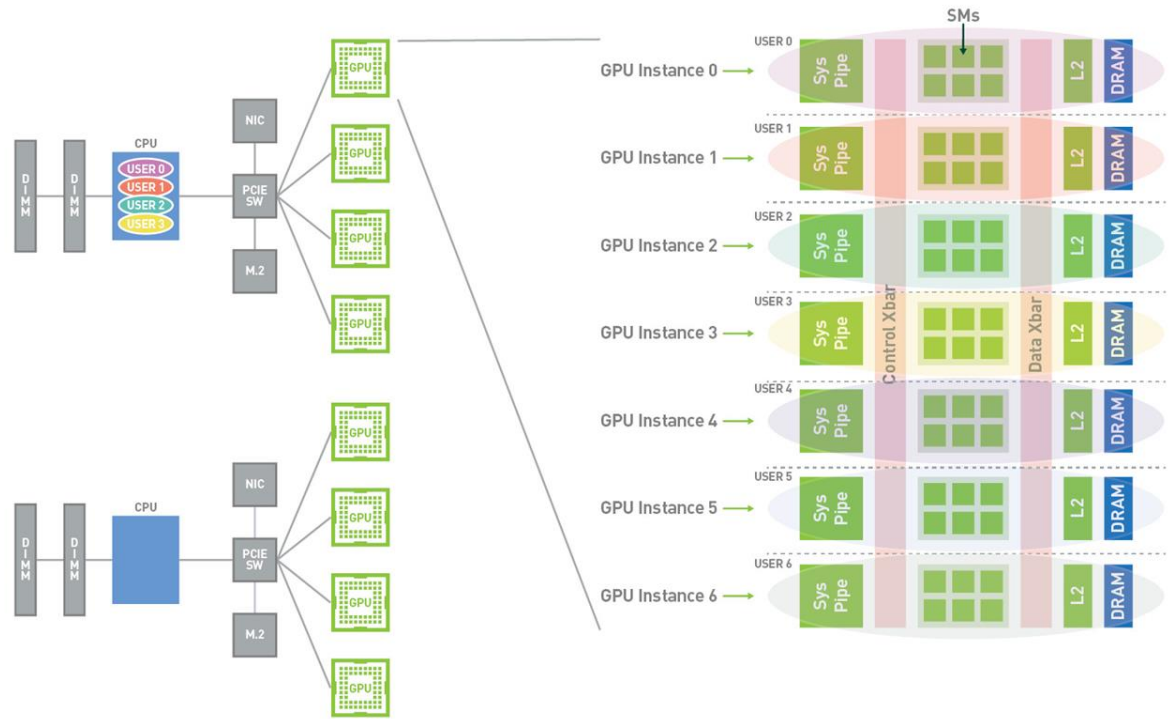
    steps:
      - name: connect and pull
        run: ssh ${ secrets.SSH_USER }}@${ secrets.SSH_HOST }} "cd bot && git pull && make deploy"
      - name: cleanup
        run: rm -rf ~/.ssh
```

Docker

- + Воспроизводимая среда
- + Можно собирать и использовать в CI
- + Поддержка MIG
- + Docker-compose

MIG

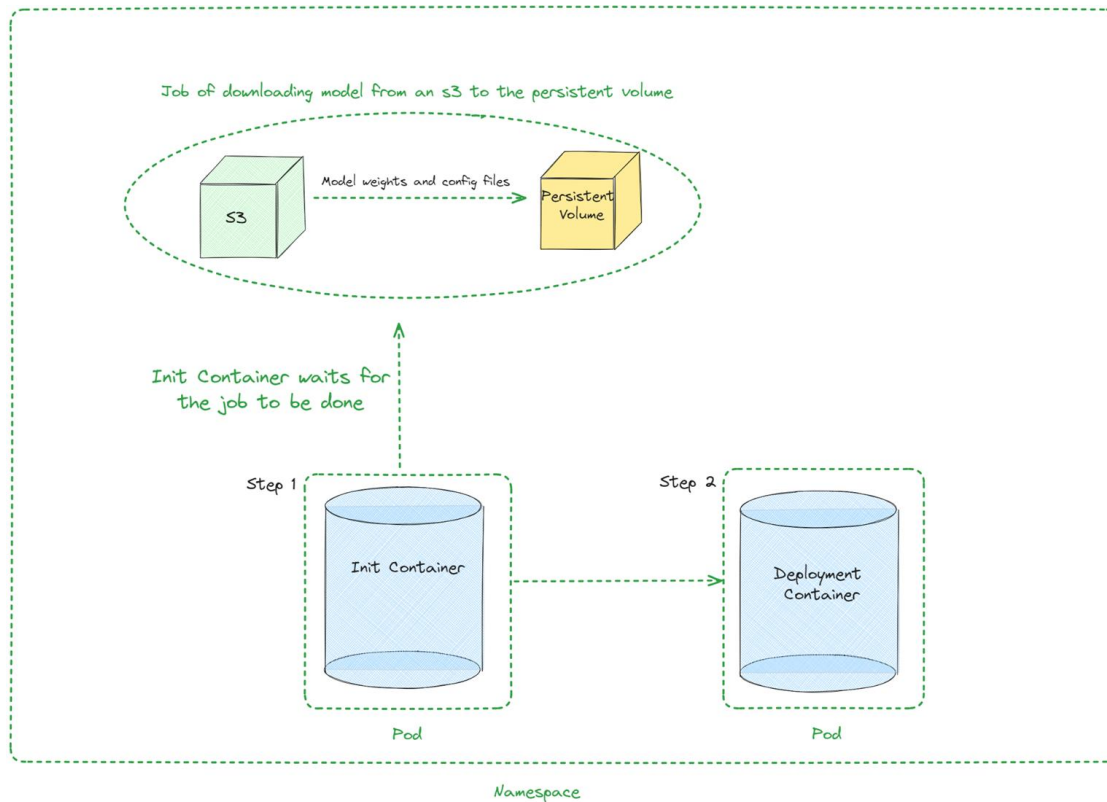
MULTI-INSTANCE GPU ("MIG")



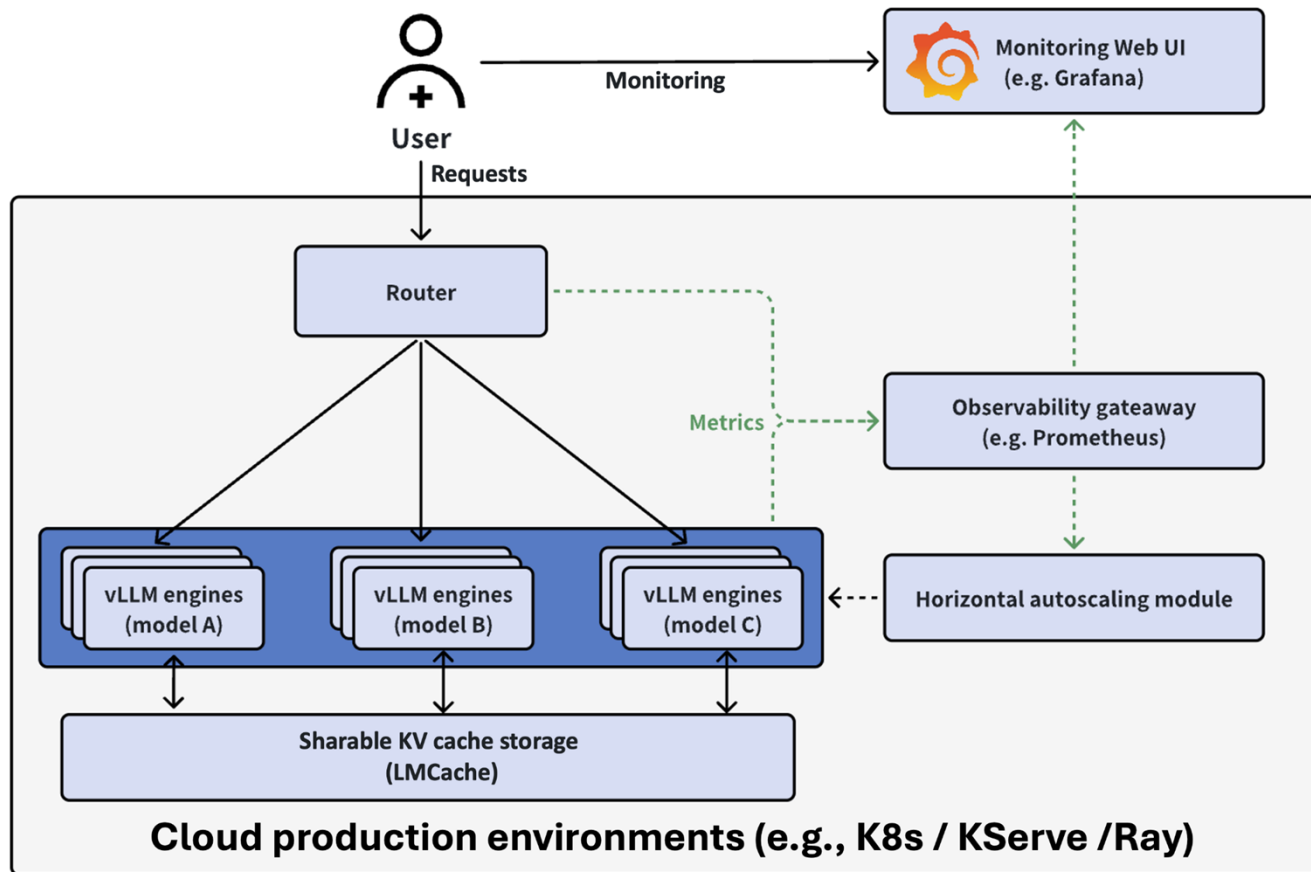
K8s

- + Продвинутый мониторинг
- + Гибкое управление ресурсами
- + Настраиваемый dns
- Сложно настраивать

Пример деплоя с vLLM



vLLM Production Stack



Monitoring & observability

Prometheus & Grafana



Observability

- Langfuse
- Langsmith
- Logfire
- ...

Go to...

x K

Home

Dashboards

Observability

Tracing

Sessions

Users

Prompt Management

Prompts

Playground

Evaluation

Scores

LLM-as-a-Judge

Human Annotation

Datasets

Settings

Support

FK Felix Krauth
felix@langfuse.com

Trace: eb346ba8-19d9-4368-9b71-022cd17f25d7

with-context



Search (type, title, id)

Timeline

qa



11.58s \$0.000371

retrieval

1.55s \$0.000001

prompt-embedding DEBUG

0.67s 12 → 0 (\$12) \$0.000001

vector-store

0.69s

context-encoding

fetch-prompt-from-langfuse

0.47s

generation

9.57s 1301 → 291 (\$1592) \$0.00037

qa ID

+ Add to datasets

Annotate

1



2025-07-16 23:44:16.449

Session: lf.docs.conversation.s7A0W7i

User ID: u-M9QTUzj

Env: default

Latency: 11.58s

Total Cost: \$0.000371

1,313 → 291 (\$1,604)

Release: 044d0162fe3dbddd09f88a554521ddf73726618c

Preview Scores

Formatted JSON

Input

View as markdown

"How to update the input and output in langgraph traces?"

Output

View as markdown

"To update the input and output in Langfuse traces, it's essential to understand that Langfuse allows meticulous tracking of steps within the AI process by logging both inputs and outputs of large language models (LLMs). This means that every interaction you have can be captured and displayed transparently within the UI.

Here's how you can effectively manage and visualize these updates:

Steps to Update Traces

1. **Log Inputs and Outputs**: Ensure you're logging the inputs and outputs of

...expand (989 more characters)

"

Metadata



```
{  
  "pathname": "/docs/demo"  
}
```


Спасибо за внимание.